# Sabir Saluja

sabsaluja@montefiore.org

## Education

**University of Southern California** — Los Angeles, CA
Bachelor of Science (B.S.), Neuroscience — Aug 2014 - May 2018

**Renaissance School of Medicine at Stony Brook University** — Stony Brook, NY
Doctor of Medicine (M.D.) — May 2020 - Dec 2024

## Experience

**Montefiore Einstein Medical Center** — Bronx, NY
Residency, Diagnostic Radiology — July 2026 - June 2030

**Sutter Health Alta Bates Summit Medical Center** — Oakland, CA
Internship, Transitional Year — July 2025 - June 2026

**Stanford Medicine, Department of Neurosurgery** — Stanford, CA
Research Fellowship — Sep 2024 - Present

- PI: Dr. Ashwin Ramayya, MD, PhD
- Leading investigations on neurosurgical interventions for chronic pain.
- Completed meta-analysis of placebo interventions for facial pain, in submission.
- Lead imaging analysis and writing a grant for focused ultrasound to treat cancer pain.
- Exam patients in TBI clinic, assist with data acquisition from behavioral tasks.

**Stony Brook Medicine, Department of Neurosurgery** — Stony Brook, NY
Research — May 2021 - Present

- PI: Dr. Charles Mikell III, MD and Dr. Sima Mofakham, PhD
- Investigating thalamo-prefrontal circuits involved in the recovery of command following severe TBI.
- Designed neuroimaging pipeline that is used by the lab for analyzing all structural and diffusion MRI data. Helped implement a new high-resolution DWI sequence optimized for TBI patients.
- Mentored underclassmen in basic programming for data analysis and interpretation of neuroimaging data for research.

**Penn Medicine, Department of Neurosurgery** — Philadelphia, PA
Research — Sep 2021 - Sep 2024

- PI: Dr. Casey Halpern, MD
- Tractography-based targeting for SEEG-guided DBS for medically refractory OCD (NCT05623306)
- Designed neuroimaging pipeline that is used by the lab for analyzing all MRI data.
- Investigating outcomes of DBS for PD, with a focus on non-motor effects of stimulation.

**Stanford Medicine, Department of Neurosurgery** — Stanford, CA
Research — May 2018 - May 2020

- PI: Dr. Casey Halpern, MD
- Designed protocols for analysis of neuroimaging data for neuromodulation procedures such as deep brain stimulation, MRI-guided focused ultrasound (MRgFUS).
- Assisted with collecting patient data from experimental neuromodulatory trials for loss of control binge-eating, OCD, and MRgFUS for essential tremor.

**USC Institute for Neuroimaging and Informatics** Los Angeles, CA
Research Aug 2015 - Aug 2017

- PI: Dr. Kristi Clark, PhD
- Developed novel techniques for the analysis and integration of MRI with genomic data.
- Completed Neuroscience Honors Thesis for NEUR-490 (Directed Research), Provost Undergraduate Research Fellowship in Summer 2016

**Transcriptic** Menlo Park, CA
Internship (Bioengineering) May 2015 - Aug 2015

- Summer internship at an early-stage, Y Combinator-backed startup.
- Assisted in development and quality control of robotic platform to automate biology experiments from the cloud.
- Refactored command line interface for experiment submission. Optimized job scheduling on finite hardware resources using Simplex optimization.

## Publications

1. **Saluja**, **S.**, Qiu, L., Wang, A. R., Campos, G., Seilheimer, R., McNab, J. A., Haber, S. N., Barbosa, D. A., and Halpern, C. H., "Diffusion MRI tractography guides investigation of the zona incerta: A novel target for deep brain stimulation", *Biological Psychiatry*, S0006322324011053, Feb. 2024, ISSN: 00063223.

2. **Saluja**, **S.**, Robertson, J., Saadon, J. R., Lubin, N., Huang, C., Saalmann, Y., Mikell, C. B., and Mofakham, S., "Diffusion MRI tractography of central thalamic projections characterizes the recovery of command-following and arousal in severe traumatic brain injury (submitted)", Sep. 2024.

3. Gammel, T. C., Alkadaa, L. N., Saadon, J. R., **Saluja**, **S.**, Servider, J., Cleri, N. A., Egnor, M., Davis, R. P., Huang, C., Saalmann, Y. B., Mofkaham, S., and Mikell, C. B., "Brain circuitry of consciousness: A review of current models and a novel synergistic model with clinical application", *Neurosurgery Practice*, vol. 4, 2023.

4. **Saluja**, **S.**, Razzaq, B., Servider, J. A., Seidman, R., and Mushlin, H., "Diffuse midline glioma, h3k27-altered, of the conus medullaris presenting as acute urinary retention: Illustrative case", *Journal of Neurosurgery: Case Lessons*, vol. 5, no. 16, CASE22529, Apr. 17, 2023, ISSN: 2694-1902.

5. **Saluja**, **S.**, Saadon, J., Cosgrove, M., Zhang, J., Huang, C., Mikell, C. B., and Mofakham, S., "Diffusion-weighted MRI tractography of midline thalamic subnuclei characterizes the return of consciousness following severe traumatic brain injury", *Neurosurgery*, vol. 69, pp. 37–38, Apr. 2023, ISSN: 0148-396X, 1524-4040.

6. Barbosa, D. A. N., Kuijper, F. M., Duda, J., Wang, A. R., Cartmell, S. C. D., **Saluja**, **S.**, Cunningham, T., Shivacharan, R. S., Bhati, M. T., Safer, D. L., Lock, J. D., Malenka, R. C., Oliveira-Souza, R. de, Williams, N. R., Grossman, M., Gee, J. C., McNab, J. A., Bohon, C., and Halpern, C. H., "Aberrant impulse control circuitry in obesity", *Molecular Psychiatry*, vol. 27, no. 8, pp. 3374–3384, Aug. 2022, ISSN: 1359-4184, 1476-5578.

7. Cosgrove, M. E., Saadon, J. R., Mikell, C. B., Stefancin, P. L., Alkadaa, L., Wang, Z., **Saluja**, **S.**, Servider, J., Razzaq, B., Huang, C., and Mofakham, S., "Thalamo-prefrontal connectivity correlates with early command-following after severe traumatic brain injury", *Frontiers in Neurology*, vol. 13, p. 826 266, Feb. 18, 2022, ISSN: 1664-2295.

8. Kakusa, B., Huang, Y., Barbosa, D. A., Feng, A., Gattas, S., Shivacharan, R., Lee, E. B., Kuijper, F. M., **Saluja**, **S.**, Parker, J. J., Miller, K. J., Keller, C., Bohon, C., and Halpern, C. H., "Anticipatory human subthalamic area beta-band power responses to dissociable tastes correlate with weight gain", *Neurobiology of Disease*, vol. 154, p. 105 348, Jul. 2021, ISSN: 09699961.

9. Kakusa, B., **Saluja**, **S.**, Barbosa, D. A., Cartmell, S., Espil, F. M., Williams, N. R., McNab, J. A., and Halpern, C. H., "Evidence for the role of the dorsal ventral lateral posterior thalamic nucleus connectivity in deep brain stimulation for gilles de la tourette syndrome", *Journal of Psychiatric Research*, vol. 132, pp. 60–64, Jan. 2021, ISSN: 00223956.

10. Feng, A. Y., Wong, S., **Saluja**, **S.**, Jin, M. C., Thai, A., Pendharkar, A. V., Ho, A. L., Reddy, P., and Efron, A. D., "Resection of olfactory groove meningiomas through unilateral vs. bilateral approaches: A systematic review and meta-analysis", *Frontiers in Oncology*, vol. 10, p. 560 706, Oct. 22, 2020, ISSN: 2234-943X.

11. Kakusa, B., **Saluja**, **S.**, Dadey, D. Y. A., Barbosa, D. A. N., Gattas, S., Miller, K. J., Cowan, R. P., Kouyoumdjian, Z., Pouratian, N., and Halpern, C. H., "Electrophysiology and structural connectivity of the posterior hypothalamic region: Much to learn from a rare indication of deep brain stimulation", *Frontiers in Human Neuroscience*, vol. 14, p. 164, May 15, 2020, ISSN: 1662-5161.

12. **Saluja**, **S.**, Barbosa, D. A. N., Parker, J. J., Huang, Y., Jensen, M. R., Ngo, V., Santini, V. E., Pauly, K. B., Ghanouni, P., McNab, J. A., and Halpern, C. H., "Case report on deep brain stimulation rescue after suboptimal MR-guided focused ultrasound thalamotomy for essential tremor: A tractography-based investigation", *Frontiers in Human Neuroscience*, vol. 14, p. 191, Jun. 26, 2020, ISSN: 1662-5161.

13. Kakusa, B., **Saluja**, **S.**, Tate, W. J., Espil, F. M., Halpern, C. H., and Williams, N. R., "Robust clinical benefit of multi-target deep brain stimulation for treatment of gilles de la tourette syndrome and its comorbidities", *Brain Stimulation*, vol. 12, no. 3, pp. 816–818, May 2019, ISSN: 1935861X.

14. Singh, A., Agrawal, S., Gargya, S., **Saluja**, **S.**, Kumar, A., Kumar, A., Kalra, K., Thind, M., Saluja, S., Stone, L., Ali, F., Duarte-Chavez, R., Marchionni, C., Sholevar, F., Shirani, J., and Nanda, S., "Posttraumatic stress disorder after myocardial infarction and coronary artery bypass grafting", *International Journal of Critical Illness and Injury Science*, vol. 7, no. 2, p. 84, 2017, ISSN: 2229-5151.

## Oral Presentations

1. **Saluja, S.**, Cosgrove, M., Saadon, JR., Zhang, J., Huang, C., Mikell, C., Mofakham, S. Diffusion-weighted MRI tractography of midline thalamic subnuclei characterizes the return of consciousness following severe traumatic brain injury. Congress of Neurological Surgeons 2022 Annual Meeting. Oct. 11, 2022. San Francisco, CA.

2. **Saluja, S.**, Kakusa, B., Barbosa, DAN., Dadey, D., Halpern, CH. Electrophysiology and Structural Connectivity of the Posterior Hypothalamic Region in a Case of Deep Brain Stimulation for Concurrent Management of Cluster Headache and Essential Tremor. World Society for Functional and Stereotactic Neurosurgery. Jun. 25, 2019. New York, NY.

## Volunteering

**Stony Brook Home (SB HOME)** Aug 2020 - Present
Student-run free clinic serving patients in greater Suffolk County area.
Perform clinical exams, assist with phlebotomy, write visit notes. Teach underclassmen techniques in physical exams and communication.

**Student National Medical Organization** Aug 2020 - Present
Mentor underserved undergraduate students on medical school admissions advice.
Meet monthly with students, answer questions about path to medical school and provide help.

**Lava Mae** Feb 2019 - Feb 2020
Helped run a free shower service in the Mission district of San Francisco, CA.
Assisted with shower cleaning and maintenance, scheduling, and health outreach initiatives.